

Dr. Thomas E. Kent

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Machine learning leader with 10+ years' experience spanning applied research and delivery leadership in machine learning, autonomous systems, robotics in defence and national security. Ensures technical direction and scientific standards across multi-programme portfolios, line manages senior technical staff, and has led or contributed to bids winning over £2.5m in new and follow-on work. Combines deep technical expertise in machine learning, optimisation, simulation, decision-making, control, and multi-agent systems with strong people leadership, hiring, and customer engagement, translating ambiguous operational problems into deployed solutions.

- 10+ years' applied ML leadership in autonomous systems, robotics, and defence & national security.
- Led or contributed to bids winning over £2.5m in new and follow-on work.
- Experienced manager of teams and people, currently line managing six senior technical staff.
- 9 first-author publications from PhD and postdoctoral research.

LEAD MACHINE LEARNING SCIENTIST | MIND FOUNDRY | 2023 - PRESENT

- Set technical direction and scientific standards across Mind Foundry's Defence and National Security portfolio, ensuring machine learning research and engineering delivery are aligned with customer needs, product strategy, and company objectives.
- Lead cross-functional teams of Machine Learning Scientists and Engineers delivering applied ML solutions for Defence and National Security customers.
- Line manage six senior individual contributors, including managers, covering performance, development, mentorship, organisational design, and team growth within an evolving team structure.
- Work closely with commercial, product, and delivery stakeholders on go-to-market activities, including opportunity development, customer engagement, use-case discovery, ML solution ideation, and proposal development for new and follow-on work.
- Partner directly with customers and internal stakeholders to translate ambiguous operational challenges into actionable ML solutions, technical roadmaps, and delivery plans.
- Lead a large number of technical bids and proposals; helped win over £2.5m in new and follow-on contract value.
- Design and run hiring, onboarding, mentoring, and development of technical talent, helping build a growing number of high-performing ML experts across multiple domains.
- Hands-on Technical Lead across multiple ML projects, setting technical direction while directly contributing to the development and maturation of novel ML solutions from early-stage research through to real-world deployment.
- Projects spanning computer vision, anomaly detection, drift monitoring and model lifecycle management, predictive maintenance, electronic surveillance, recommendation systems, localisation and tracking.

MACHINE LEARNING SCIENTIST | MIND FOUNDRY | 2022 - 2023

- Developed machine learning, signal processing, simulation, and modelling techniques for novel solutions across domains including radar and Radio Frequency sensing.
- Technical lead on projects: architecting solutions, setting technical direction, and running delivery in partnership with customers and domain experts.
- Delivered projects spanning radar, tracking, multi-agent systems, explainability, and interpretability.
- Significantly involved in writing successful bids for new projects and follow-on work on active engagements.

SENIOR RESEARCH ASSOCIATE | MATHEMATICS & COMPUTER SCIENCE | UNIVERSITY OF BRISTOL | 2018 – 2022

- Led applied research into heterogeneous multi-agent systems: tasking, routing, and surveillance under the Thales-Bristol partnership in hybrid autonomous systems engineering project.
- Designed a decentralised evolutionary algorithm to solve a dynamic task allocation and routing problem.
- Created scalable single-agent policies for persistent surveillance, combining reinforcement learning, neuro-evolution, and heuristic approaches, exploring emergent behaviour when scaling to multiple agents.
- Built multi-agent system simulation software for testing and deployment of novel algorithms across a range of vehicles, including a flexible web-based GUI for human-in-the-loop scenarios.
- Supervised six PhD and Masters students.

RESEARCH ASSOCIATE | BRISTOL ROBOTICS LABORATORY | UNIVERSITY OF BRISTOL | 2015 – 2018

- Developed path-planning and decision-making capabilities for the flagship £5m UK driverless car project.
- Scoped, implemented, and tested algorithms to reliably and repeatably produce high-level trajectories for a series of increasingly complex real-world, participant-led trials.
- Designed a Behaviour Tree approach for flexible, modular construction of trials, paired with a suite of path-planning algorithms delivering comfortable, safe trajectories across scenarios; built to run on both the real vehicle and the project's driving simulator.
- Successfully drove sixty participants through scenarios involving pedestrians, cyclists, and cars, demonstrating technology including vehicle-to-vehicle communication.

EDUCATION

PhD, Aerospace Engineering | University of Bristol | 2011-2015

- **Thesis:** *'Optimal Routing and Assignment for Commercial Formation Flight'*
- Developed a scalable analytic-geometric approach to commercial formation flight routing, enabling fast cost estimation and large-scale fleet assignment via Mixed Integer Linear Programming.
- Demonstrated fuel savings of 8–13% in a Transatlantic case study, with consistent gains shown across flights of varied distance, geography, and formation size.

MA (Hons), Pure Mathematics | University of Edinburgh | 2007-2011

SELECTED PUBLICATIONS

4 of 9 first-author publications from PhD and postdoctoral research; full list at tomekent.com/publications.

- Kent, T. E. & Richards, A. G. Analytic Approach to Optimal Routing for Commercial Formation Flight. *Journal of Guidance, Control, and Dynamics*, 38, 1872–1884 (2015).
- Kent, T., Pipe, A., Richards, A., Hutchinson, J. & Schuster, W. A Connected Autonomous Vehicle Testbed: Capabilities, Experimental Processes and Lessons Learned. *Automation*, 1, 17–33 (2020).
- Kent, T. E. & Richards, A. G. Decentralised Multi-Demic Evolutionary Approach to the Dynamic Multi-Agent Travelling Salesman Problem. *Genetic and Evolutionary Computation Conference (GECCO '19)*, 147–148 (2019).
- Kent, T. E., Richards, A. G. & Johnson, A. Single-Agent Policies for the Multi-Agent Persistent Surveillance Problem via Artificial Heterogeneity. *European Conference on Multi-Agent Systems (EUMAS 2020)*.

SKILLS

Machine Learning | Optimisation | Simulation | Signal Processing | Multi-Agent Systems | Decision-Making
Python | PyTorch | sklearn | Docker | Git | Air-Gapped Deployment | Local LLMs & RAG
Technical Delivery | People Leadership | Hiring | Bid Writing | Customer & Stakeholder Engagement